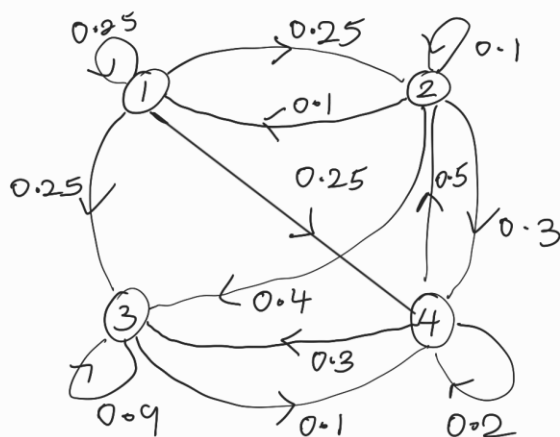
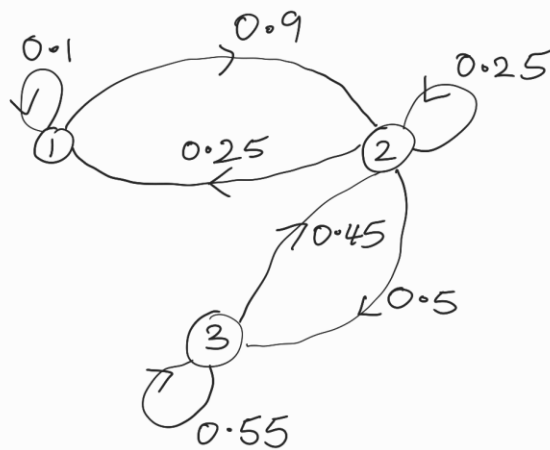
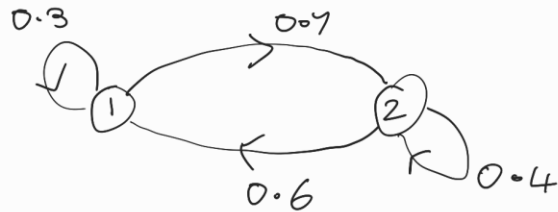


Problem Set 10: Markov Chains

Q1 a) $\begin{pmatrix} 0.1 & 0.9 \\ 0.8 & 0.2 \end{pmatrix}$; b) $\begin{pmatrix} 0 & 0.75 & 0.25 \\ 0.3 & 0.3 & 0.4 \\ 0.85 & 0.1 & 0.05 \end{pmatrix}$; c) $\begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 0.3 & 0.1 & 0.3 & 0.3 \\ 0.2 & 0 & 0 & 0.8 \\ 0 & 0.2 & 0.7 & 0.1 \end{pmatrix}$

Q2



Q3 b) $\mathbb{P} = \begin{pmatrix} 0.1 & 0.9 \\ 0.8 & 0.2 \end{pmatrix}$ c) 0.28

Q4 b) Order: 1,2,3 = Shaggy, Scooby Doo, Fred; $\mathbb{P} = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \\ 3/4 & 1/4 & 0 \end{pmatrix}$ c) 0.25

d) Shaggy: $x_1 = 0.4286$, Scooby Doo: $x_2 = 0.2857$, Fred: $x_3 = 0.2857$

Q5

b) Order 1, 2, 3 = Chinese, Pizza, Home

$$\mathbb{P} = \begin{pmatrix} 0.2 & 0.2 & 0.6 \\ 0.1 & 0.2 & 0.7 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}$$

c) 0.12

d) Chinese: $x_1 = 0.11$, Pizza: $x_2 = 0.12$, Home: $x_3 = 0.73$